# AlphaLink CAM Reference Manual

Oct 13, 2026 · @Eduard

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# Introduction

AlphaLink CAM is a drawing and machining program made only for the Harrison Alpha 400T lathe with its FANUC Power Mate 0 control and the Harrison T macros. It replaces the DOS program ALPHA-T (ALPHALINK V1.2): it writes the same programs, byte for byte, and sends them over the same serial cable.

This manual follows the order of the ALPHA-T manual (LNK/T-G1.2, 2002): the screen, then every drawing (CAD) command, every machining (CAM) command, checking the work, files, sending to the machine, the tool library and problems. Where ALPHA-T had a one-letter key (*Heisse Taste*), it is given with the command; those keys work the same here. For a step-by-step introduction, use the Training Manual first.

Conventions:

- **Bold** names are buttons, menu items and fields as the app shows them in English. The Ukrainian tab uses the Ukrainian names.
- X is always a **diameter**, Z a length; Z0 is the front face and the part lies in -Z.
- **Tap** means a finger, pen or mouse click. **Enter** accepts, **Esc** cancels, as in ALPHA-T.
- Feeds are always per revolution in the drawing's units (mm or inch).

# Starting and the screen

## Starting

AlphaLink CAM is installed as the Flatpak `io.github._0wn3r.AlphaLinkCAM`; a `.flatpak` bundle installs with `flatpak install --user <file>`.

Start **AlphaLink CAM** from the applications menu, or `flatpak run io.github._0wn3r.AlphaLinkCAM`. Without a part open the window offers **New Part…** and **Open…**, and **Open the Training Manual** under them.

## A new drawing

**New Part…** (Ctrl+N) asks what ALPHA-T asked when starting a drawing, plus two display options:

| Field | Meaning |
|---|---|
| Units | mm or inch for the drawing |
| Billet diameter | the bar's diameter (X) |
| Billet length | how far the bar sticks out of the chuck (Z) |
| Bore | inner diameter of a tube billet; 0 = solid bar |
| Chuck jaws | on the outside (as ALPHA-T draws them) or in the bore |
| Show tailstock | draws the tailstock centre at the front face |

The tool change position is X100 Z50 and the first tool comes from **Preferences**; both can be changed later with a **Tool Change** operation. **Part Settings** in the main menu changes the billet, jaws and tailstock of an open part.

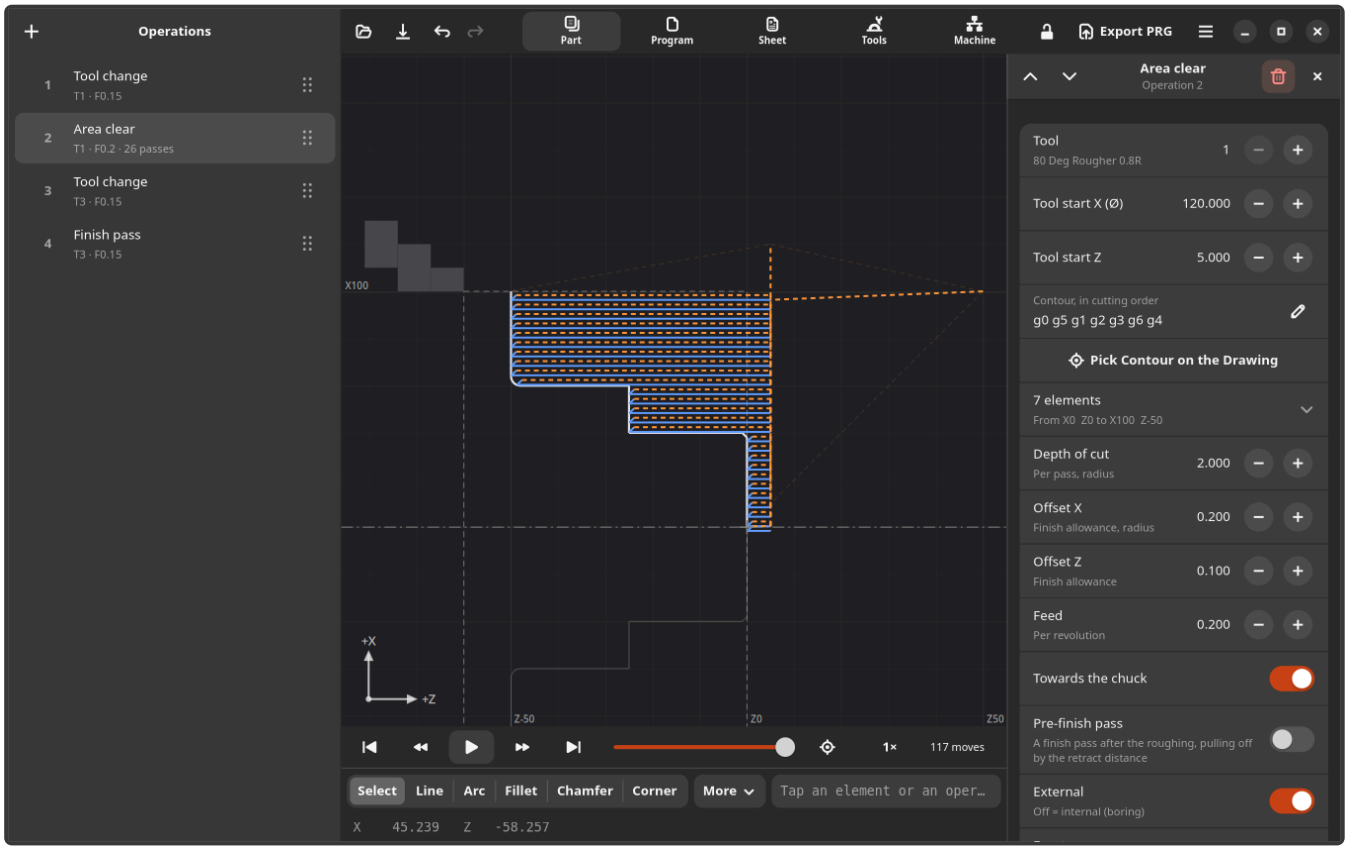
## Screen layout

| Area | What it holds |
|---|---|
| Header bar | open, save, undo/redo, page switcher, touch lock, **Export PRG**, main menu |
| Operations list (left) | the machining steps in order; **+** adds one |
| Drawing (Part page) | billet, chuck jaws, part, toolpaths, grid, axis arrows (+Z right, +X up), the cross at X0 Z0 |
| Simulation bar | first move, previous pass, play/pause, next pass, show all, playback speed, start of the selected operation; move, operation, pass and program line |
| Tool bar | Select, Line, Arc, Fillet, Chamfer, Corner, **More**, the `X,Z` entry, **Done** |
| Coordinate readout | the pointer's X and Z |
| Operation editor (right) | the settings of the selected operation |

Pages: **Part** (drawing and toolpaths), **Program** (the program as sent), **Sheet** (setup sheet), **Tools** (tool library), **Machine** (sending and backup). On a narrower window the screen folds in steps. Below 1270 px the operations list hides behind the Operations button at the top left and slides in over the drawing. Below 1150 px the page switcher moves to the

bottom. Below 1010 px (half a Surface, or a Surface held upright) the editor slides in over the drawing too, and open, save and undo move to the main menu. Below 700 px the editor fills the screen.

CAD and CAM are not separate screens as in ALPHA-T: draw with the tool bar, machine with the operations list, on the same page. ALPHA-T's **C** key opens the operations menu.




## Drawing commands (CAD)

Only the upper half of the part is drawn; the faint lower half is its mirror image. Coordinates are typed in the entry under the drawing as `X,Z` (absolute) or `@dX,dZ` (from the last point), then Enter, or set by tapping: taps snap to line ends, the axis and the grid, and lock to Z, X or 45°. Tapping the first point again closes the profile and ends the line. With a finger, a chip on the last line shows where it ends; tapping it opens the keypad to correct that point.

**The cursor.** As in ALPHA-T there is a cursor (a cross): the pointer's last place on the drawing, or where the keys below put it. Arrow keys move it by the step (start 10 mm; PgUp/PgDn ×10 / ÷10, down to 0.001), Home puts it in the middle of the view, End on X0 Z0, and Enter takes it as the next point.

## Finding

| Command | Key | What it does |
|---|---|---|
| Find point | P | cursor to the nearest line end, arc end or arc centre |
| Find intersection | I | cursor to the nearest crossing of two elements |
| Find line | L | marks the nearest line; then - or Delete erases it |
| Find arc | A | marks the nearest arc; then - or Delete erases it |
| Cursor to X,Z | G | asks X,Z; the cursor goes there (also **More ▸ Move Cursor to X,Z**) |
| Cursor by dX,dZ | H | asks an increment (also **More ▸ Move Cursor by dX,dZ**) |
| Cursor by length, angle | Y | asks length,angle (0° towards -Z, 90° towards +X; also **More ▸ Move Cursor by Length and Angle**) |
| Previous / next element | [ / ] | selects the previous or next element along the profile |

## Lines and arcs

| Command | Key | What it does |
|---|---|---|
| Start point | S | starts a line at the cursor |
| Line X,Z | X | the Line tool, ready for typed points (typing a number does the same) |
| Line length, angle | D | asks length,angle from the last point (also **More ▸ Draw Line by Length and Angle**) |
| Line by taps | + | the Line tool for tapping points; **Done** or Esc ends the line |
| Arc (3 points) | | **Arc** tool: start, a point on the arc, end |
| Circle | | **More ▸ Circle…**: centre X, Z and radius; drawn as two halves split at the 3 o′clock point |
| Tangent | | **More ▸ Tangent**: a point (or line end) and an arc, or two arcs; tap the wanted line among the dashed ones |

| Command | Key | What it does |
|---|---|---|
| Tangent at an angle | | **More ▸ Tangent at an Angle…**: an arc and an angle; 1.5 × long |
| Parallel line | | **More ▸ Parallel Line…**: a line and a distance; 1.5 × as long, either side |
| Blend arc | O | **More ▸ Blend Arc…**: an arc of a radius touching two points, lines or arcs |

## Corners and editing

| Command | Key | What it does |
|---|---|---|
| Radius / chamfer at a corner | F | the corner nearest the cursor: radius (Fillet) or chamfer size × angle (Chamfer) |
| Fillet tool | — | tap corners for radii |
| Trim corner | K | **Corner**: two lines are extended or trimmed until they meet |
| Split arc | | **More ▸ Split Arc**: tap where the arc should be split |
| Mirror | | **More ▸ Mirror Image**: the whole drawing copied mirrored about the spindle axis at once, then "Is this correct?" (No takes it back), as in ALPHA-T |
| Erase element | Delete | erases the selected line or arc, anywhere on the Part page outside text fields (Undo brings it back) |
| Erase drawing | Q | erases everything after asking (also **More ▸ Erase Drawing…**, the last item) |
| Chamfer / round outside corners | — | More ▸ Chamfer Outside Corners… / Round Outside Corners…: one size (and angle) on every convex corner between two lines of the outer profile, as one undo step; corners whose lines are too short are left |
| Standard thread | — | More ▸ Standard Thread… on a selected diameter line: see Standard threads under Machining commands |

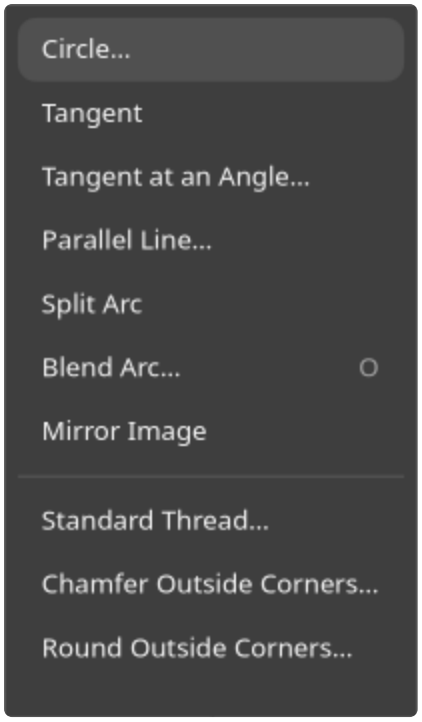
*Note:* a chamfer's size is measured along the line that runs along Z and its angle is from that line: 2×30° on a shoulder is 2 mm along the diameter, 1.155 mm down the face.

## Measuring, text and view

| Command | Key | What it does |
|---|---|---|
| Measure X,Z | M | two points; shows the X and Z distance (also **More ▸ Measure by X,Z**) |
| Measure length, angle | N | two points; shows the length and angle (also **More ▸ Measure by Length and Angle**) |
| Text | T | puts text at the cursor (not machined; also **More ▸ Add Text**) |
| Erase text | U | erases the nearest text (also **More ▸ Erase Text**) |
| Zoom | Z | zooms in at the cursor (pinch, Ctrl+wheel and Ctrl++ too) |
| Redraw / fit | R | redraws: fits the whole part (double tap, Ctrl+0) |
| Recall view | V | back to the view before |
| Operations menu | C | ALPHA-T's CAD/CAM switch: opens the list of operations |

Esc ends the current line or command and gives the keyboard back to the drawing, so the letter keys work again. Tab is not mapped (E, ALPHA-T's exit, edits here); F1 is this program's help.

**More** on the drawing bar has these commands in sections: **Construct** (Circle…, Tangent, Tangent at an Angle…, Parallel Line…, Split Arc, Blend Arc…, Mirror Image), **Features** (Standard Thread…, Chamfer Outside Corners…, Round Outside Corners…), **Cursor and Measure, a submenu** (Draw Line by Length and Angle, Move Cursor to X,Z, Move Cursor by dX,dZ, Move Cursor by Length and Angle, Measure by X,Z, Measure by Length and Angle) and **Text** (Add Text, Erase Text), then **Erase Drawing…**. With the keyboard, Enter takes the element nearest the cursor, a number dialog takes the typed value with Enter, and where there are several answers Tab moves between the dashed ones and Enter takes one.




**Editing an element** (not in ALPHA-T, which could only erase and redraw): select a line or arc and tap the pencil next to the bin, or press E. Type new start and end points (and, for an arc, a radius). With **Move connected ends too** on, the elements that met it move with it, so the profile stays joined. Undo puts it back.

A selected element shows handles at its ends: drag one to move that point, and the elements joined to it follow. With a finger a loupe shows the spot; moving slowly drags finely. A long press or a right click on an element opens **Edit Values…**, **Delete**, **Zoom Here** and **Fit Part**; the Menu key or Shift+F10 opens the same menu at the cursor. Tapping a corner where two elements meet offers both.

Three keys are added that ALPHA-T left free: **E** edits the selected element and **B** is the Chamfer tool and W the Arc tool. Hovering over a tool on the drawing bar shows its keys. **Esc** also clears the messages at the bottom of the window.

## Machining commands (CAM)

Operations are added with **+** above the operations list (or the **C** key) and are carried out in list order. Selecting one opens its editor; the arrows in its header (**Move Up**, **Move Down**, next to **Close Editor**) move it up or down, and each row's handle drags it to another place. **Duplicate Operation** and **Delete Operation** are button rows at the end of the editor; Undo brings a deleted operation back. With the operations list or the editor focused, Alt+↑ / Alt+↓ move it, Ctrl+D duplicates it and Delete deletes it. Every change regenerates the toolpaths at once. Adding Grooving, Threading or Drilling while the tool in the turret does

not suit them inserts a Tool Change to a suitable tool first. Tool warnings (a turning tool on a thread, a groove tip width that differs from the tool, an external tool inside) show as a sign on the row and on the setup sheet; they never block the program.

## Settings every operation has

- **Tool**: the tool memory slot on the control. The name from the tool library is shown under it.
- **Tool start** X, Z: where the tool goes before the operation and after each pass. Keep it clear of the billet. On the lathe the macros only start the operation with the tool within about 5 mm of this point.
- **Feed**: per revolution, any value from 0.001 up to the machine's maximum.

## Identifying the contour

Area clear, finish pass, face clear and grooving follow a **contour**: drawing elements in cutting order. A new operation takes the outer profile, without thread lines and construction lines; Grooving takes the nearest groove and its neighbours, Threading the two thread lines. To choose: **Pick Contour on the Drawing**, tap the elements in the order the tool should follow them (numbers show the order; tap again to remove; from the keyboard, Space adds or removes the element at the cursor and L, A, [ and ] move between elements), then **Done** or Enter (or 3, as in ALPHA-T). The editor lists the picked elements with their start and end points.

*Note:* as in ALPHA-T, the direction of the contour matters: it sets the direction of cutting.

## Cutting direction

| Setting | Meaning |
|---|---|
| **External** on/off | outside work, or internal (boring): sets the retract direction and the side the allowance is left on |
| **Towards the chuck** | cut towards the headstock; ALPHA-T's AWAY is refused (see Problems) |
| **Front** on/off | area clear along Z, or across the face (FACE) |

*Note:* the ALPHA-T manual says the direction the contour is picked in sets the cutting direction. Tried in ALPHA-T, a contour picked from the chuck end gives unusable programs: area clear only skims the front face and face clear makes one cut at the shoulder. The app therefore cuts area and face clear as if the contour had been picked from the front, whichever end you start at.

# Operations

| Operation | What it does | Settings | #127 |
|---|---|---|---|
| Tool Change | stops at the tool change point for the operator to fit the tool | tool, tool change point, feed | 1 |
| Area Clear | roughing passes along Z, each diameter its own pass, 45° pull-off by the retract distance (1 mm); with Follow contour each pass also cleans up the step it leaves | depth of cut (per side), offset X/Z (allowance), external, towards, front, pre-finish, follow contour, retract distance | 6 / 7 internal / 8 face |
| Pre-finish pass | a finish pass along the contour after the roughing, part of the same operation | (switch in Area Clear and Face Clear) | 10 / 11 / 12 |
| Finish Pass | one pass along the contour; approach and 0.2 mm pull-off at 45°, nose radius compensated | offset X/Z, external, face (pull-off towards the axis) | 14 / 15 internal / 16 face |
| Face Clear | roughing passes across the faces towards the axis, one depth of cut at a time along Z | depth of cut, offset X/Z, pre-finish | 8 |
| Grooving | plunges along a drawn groove, or one pass along it (finish mode); Front off = face grooving (plunges along -Z into a face) | tip width, stepover, peck depth (0 = in one go), full retract, dwell, datum left/right, finish offset X/Z, Z shift and repeats, mode area/finish, external | 9 / 17 finish |
| Threading | thread cuts between an outer and an inner diameter line | pitch, first cut depth or number of cuts, lead in (default 5), lead out, spring passes (count), external, infeed angle (0 = straight in, e.g. 30 = along one flank) | 20 / 21 spring |
| Drilling | a hole on the centre line, pecking and dwelling as set | hole bottom Z (absolute), peck (0 = in one go), dwell, retract distance (2 mm), full retract | 18 |

| Operation | What it does | Settings | #127 |
|---|---|---|---|
| Dwell | waits a number of spindle revolutions | revolutions | |
| Move / Rapid | feed or rapid move to a point | X, Z, rapid | |

## Notes

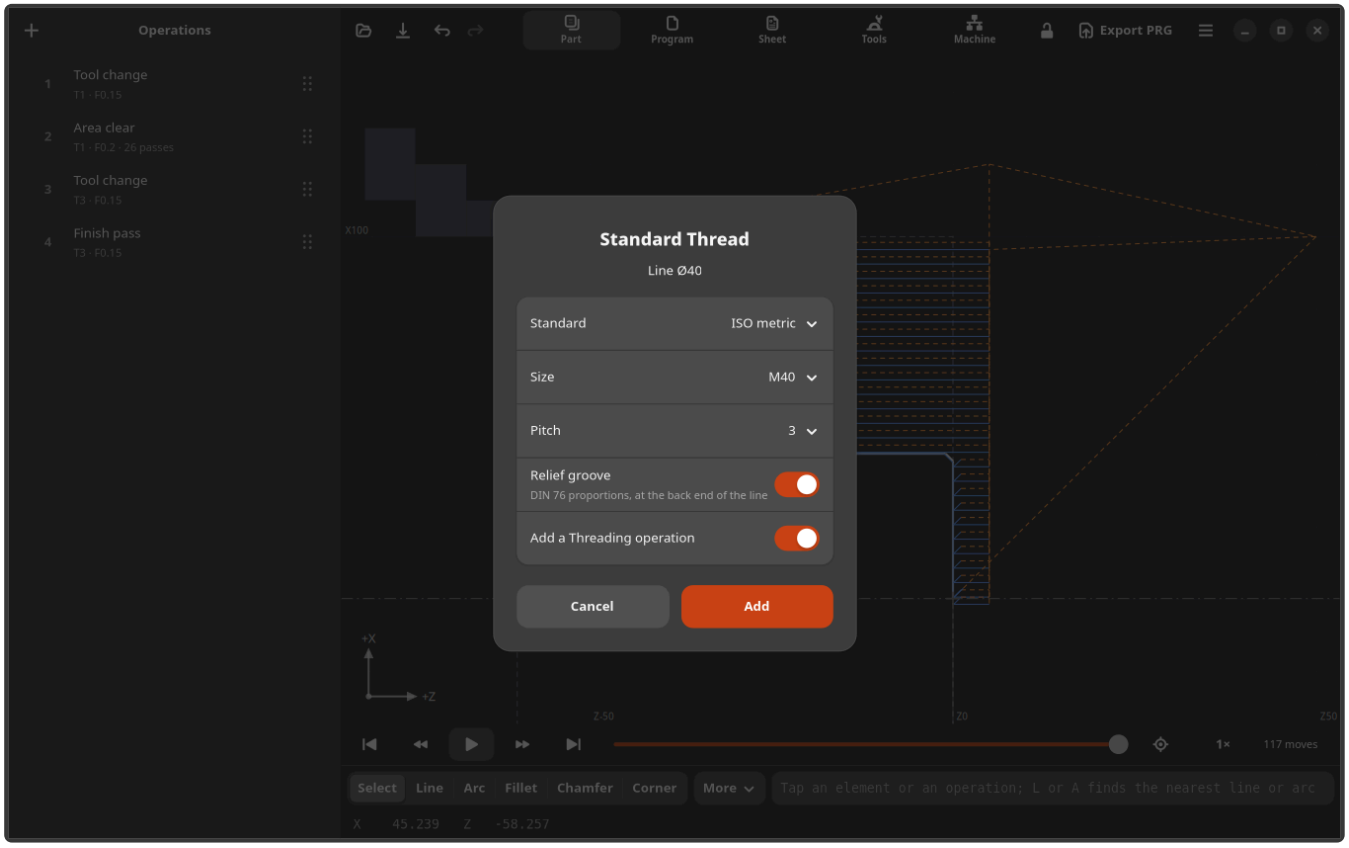
- **Depth of cut** is the true depth per side; the number of passes is worked out from it, exactly as ALPHA-T does.
- **Offset X, Z** leaves material for finishing: X on radius, Z along the axis.
- **Grooving datum** says which corner of the tip the tool memory measures; left is the -Z corner. A groove's tip width must be the real width of the tool.
- **Threading**: cut n goes to first × √n. With **Number of cuts** the first cut is depth ÷ √n. Every cut starts the lead-in in front of the thread so the slide is up to speed; the spring pass repeats the last cut. The thread's length and depth come from the two lines.
- **Dwell** and **Move / Rapid** placed after an operation run at its end, as ALPHA-T's *Verweilen*, *Vorschub zur Lichtmarke* and *Eilgang zur Lichtmarke*.
- ALPHA-T's *delete last move* and *delete all moves* are Delete Operation and Undo here; nothing is lost until you close without saving.

## Standard threads

**More ▸ Standard Thread…** threads the selected diameter line (a line along Z) in one step; with no such line selected, a banner asks you to tap it, from the ISO 261 metric or the unified inch (ASME B1.1) tables:

- **Standard**: ISO metric, UNC (inch, coarse) or UNF (inch, fine); a mm part starts on metric, an inch part on UNC.
- **Size**: M6 to M64, or 1/4″ to 2″ (UNF to 1-1/2″), the line's nearest size first. **Pitch**: for metric the coarse pitch first, then the fine ones (M40 and M50 have fine pitches only); for UNC and UNF the standard threads per inch.
- The line moves to the nominal diameter d, and the minor-diameter line is drawn under it: d − 1.22687 P for metric (depth 0.61343 P on the radius), d − 1.19078 P for UNC/UNF with the rounded UNR root (depth 0.59539 P).
- **Relief groove** (on by default) cuts an undercut with DIN 76 proportions, Ø d − 1.5 P and 3 P long, into the back end of the line; the minor line runs to its middle.
- **Add a Threading operation** adds the Threading operation with that pitch, with a Tool Change before it when the tool in the turret is not a threading tool.

An inch thread on a mm part, or a metric thread on an inch part, is converted to the part's units. The program writes the doubled pitch to 0.001 mm (0.0001 inch), so a 1/2-13 UNC on a mm part is out by less than 0.0002 mm per turn. All of it is one undo step.

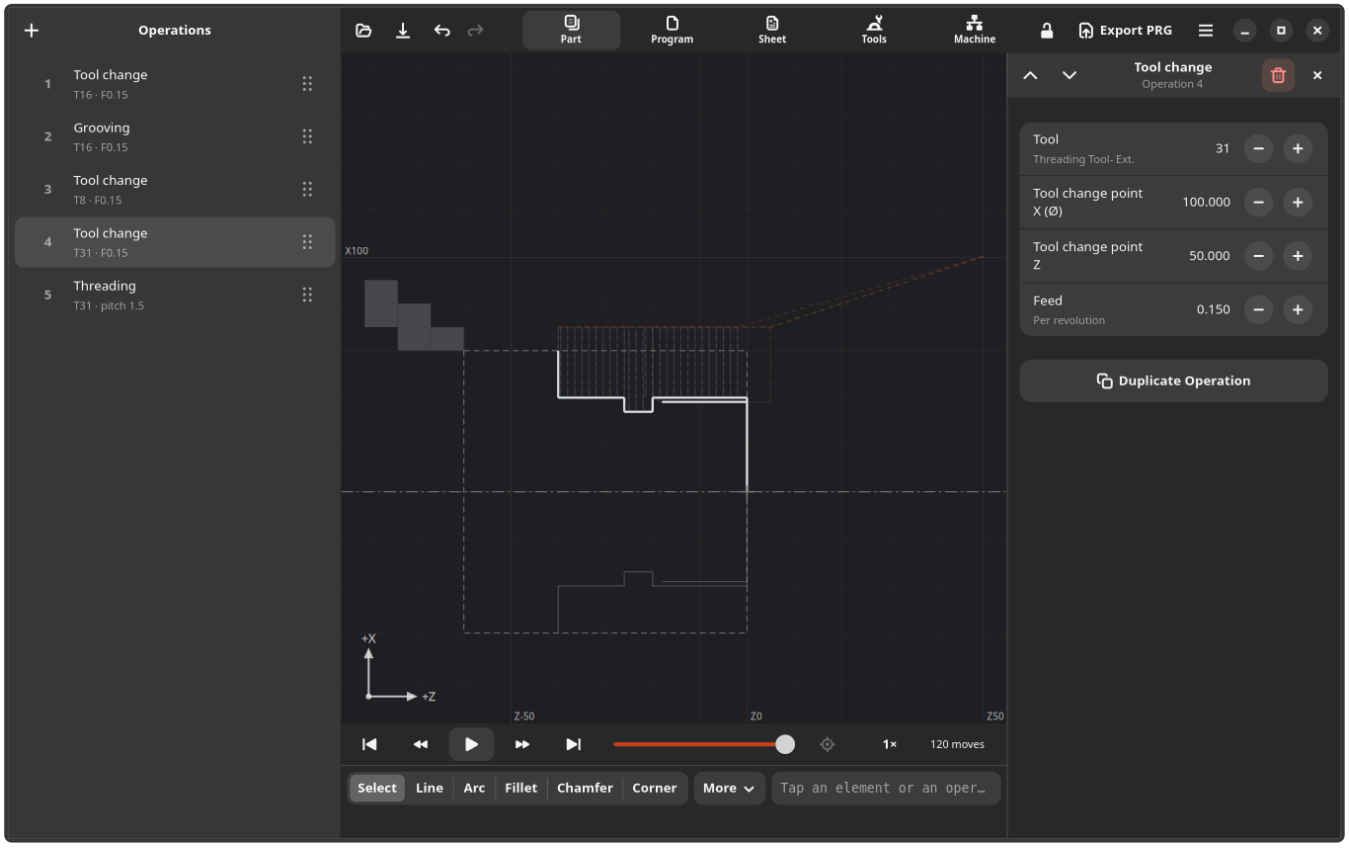



## Job templates

A template is a saved list of operations with their tools and settings but no drawing. **+ ▸ Templates** lists them; choosing one adds its operations to the part as one undo step, with the contours picked from this drawing the way a new operation picks them: the outer profile, the bore for internal operations (the chain closest to the axis), the groove and the two thread lines. Tool starts come from the billet; a drill set to go through takes the billet's length.

| Template | Operations |
|---|---|
| Shaft: rough T1, finish T3 | tool change T1, area clear (2 mm cuts, 0.2/0.1 allowance), tool change T3, finish pass |
| Bush: drill T25, bore rough and finish T12 | tool change T25, drill through, tool change T12, internal area clear, internal finish pass |
| Groove and thread: groove T16, thread T31 | tool change T16, grooving (2 mm tip), tool change T31, threading (one spring pass) |

**+ ▸ Save Operations as Template…** keeps the open part's operations as a template of your own (in ~/.config/alphalink-cam/templates). A template saved in mm is converted when it is applied to an inch part, and the other way round.




## Checking the work

Check every program before it goes to the lathe, as the ALPHA-T manual asks before saving.

- **Simulation** (under the drawing; ALPHA-T's *single step* and *animation*): ⏮ first move, ⏪ previous pass, ▶ play / pause, ⏩ next pass, ⏭ show everything. Moves not yet made are faded; the label shows the move, operation, pass and program line, and the Program page follows. Tap a toolpath to select its operation and jump there. The 1× button sets the playback speed; the button next to it jumps to the start of the selected operation. Keys: F5 plays and pauses, Ctrl+[ and Ctrl+] go to the previous and next pass, Alt+Home to the first move and Alt+End shows all; on the slider PgUp and PgDn move a tenth.
- **Hiding toolpaths**: select no operation to see all of them; select one to highlight its own.
- **Program** page: the program exactly as the lathe receives it. Tapping an operation highlights its lines. Tapping or clicking a line, Enter on the caret's line, or a search hit selects the operation that block belongs to. Pinch, Ctrl+wheel or Ctrl+= / Ctrl+− /

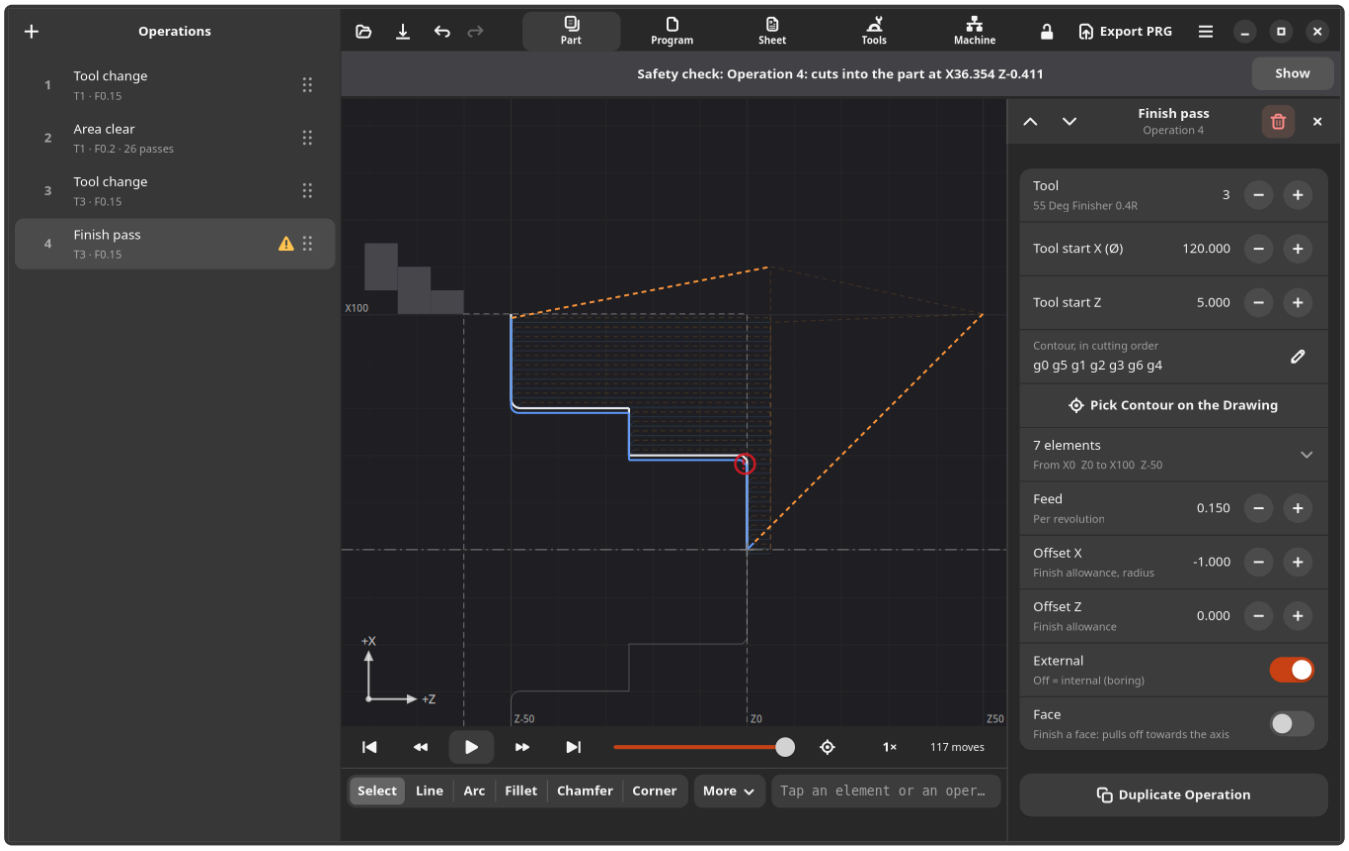
Ctrl+0 change the text size; it is remembered. The status line gives the program number, units, lines and size. Ctrl+F or the search button finds text; N10 goes to block 10 and Esc closes the search.

- **Sheet** page: the setup sheet, as ALPHA-T's print: for each operation the tool and its name, start point, limits, offsets, passes, depth of cut and feed, the block and its restart number (#140) and any tool warnings; **Tool moves** adds every move. **Save as Text…** writes it to a file.
- A **banner** at the top names any operation that cannot be made into a program and why; **Export PRG** stays off until it is fixed.

**Safety check.** Every move of the preview is checked, and a banner names the worst finding with **Show**, which zooms to it and marks it with a red ring. The operation row and the setup sheet carry the warning too. It flags:

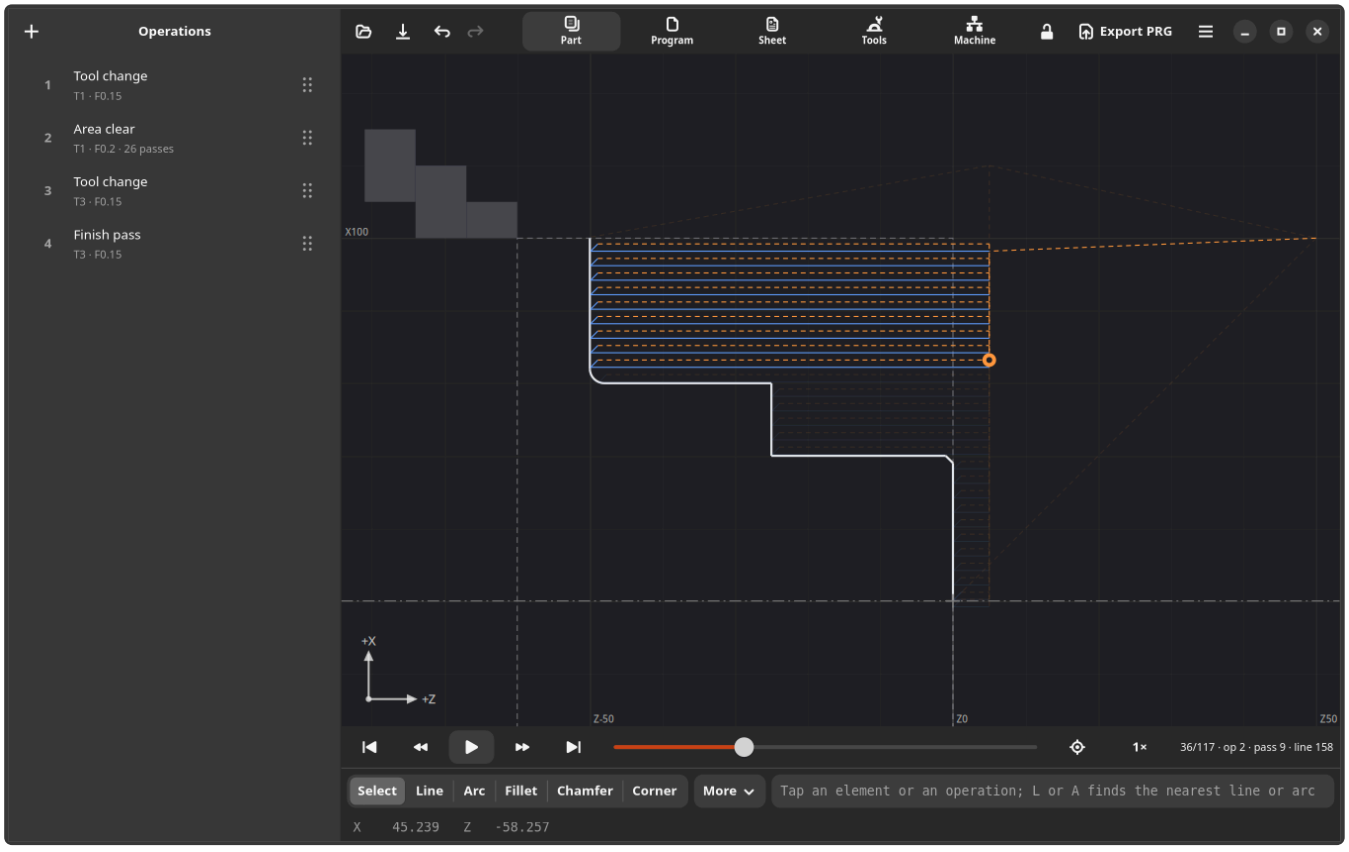
- a feed move into the finished part (the operation's own contour), or a rapid through it;
- any move into the chuck jaws (behind the billet length) or, with the tailstock shown, into its centre;
- a feed move along the contour diving steeper than the tool's front and rear clearance angles allow (ALPHA-T's Invalid tip angle).

The check finds nothing on any of ALPHA-T's sample programs. It only warns: Export PRG still works.



**Cycle time and the printable sheet.** The Sheet page estimates the cycle time per operation and in total, from every move of the program. Feed moves run at the tool's surface speed (constant surface speed, rpm = 1000 Vc / πD) up to the spindle limit; threads and drills at the rpm of their largest diameter; rapids at the rapid rate; each tool change adds the time the operator needs. Surface speed and a spindle limit can be set per tool on the Tools page (otherwise 180 m/min turning, 140 boring, 90 grooving, 60 threading and drilling); the machine's spindle limit, rapid rate and tool change time are in Preferences ▸ Cycle time. It is an estimate: acceleration and block times are left out.

**Notes for the operator** (on the Sheet page) are saved with the part. **Save as PDF…** writes a printable A4 sheet: the part with every toolpath, program number, billet and cycle time, the tools in the order they are loaded, the notes, then the sheet itself.




# File handling

| Action | Keys | What happens |
|---|---|---|
| New part | Ctrl+N | asks billet, units, bore, jaws, tailstock |
| Open | Ctrl+O | AlphaLink parts (.alcam), ALPHA-T drawings (.SDG) and DXF drawings |
| Save | Ctrl+S | writes the part (.alcam): drawing, operations, billet |

| Action | Keys | What happens |
|---|---|---|
| Save as | Ctrl+Shift+S | the same under a new name |
| Export PRG | Ctrl+E | writes the program (.PRG) and, next to it, the .SD1 listing with the same name, as ALPHA-T does |

**Main menu ▸ Save as ALPHA-T Drawing…** writes the drawing as an .SDG (with its .SDT) that DOS ALPHA-T can open; keep the name to 8 characters. With **Preferences ▸ Also save the ALPHA-T drawing (.SDG)** on, Export PRG writes it next to the program as well. An ALPHA-T drawing holds the geometry and billet only; the operations are in the .SD1.

## ALPHA-T drawings

An ALPHA-T .SDG holds only the drawing and billet; ALPHA-T keeps the operations in the .SD1 written next to it with the program. Opening an .SDG rebuilds the operations from that .SD1 and says whether the result matches the .PRG beside it. If the .SD1 cannot be found, the message offers **Choose .SD1…**.

## DXF drawings

Lines, arcs and polylines are read with the spindle axis along DXF X; the distance from the axis may be a radius (usual) or a diameter, as chosen when opening. Only the half above the axis is kept. As in ALPHA-T, CIRCLE, TEXT and blocks are skipped.

The import dialog also puts a drawing from the office in place: **Turn the drawing** (90° steps, for a part drawn upright), **Front face is on the** right or left (left mirrors it), **Spindle axis** at Y0 as drawn, in the middle (a full section) or at the lowest point (a half drawing), and **Front face on Z0**. Circles come in as two half arcs.

## Second setup

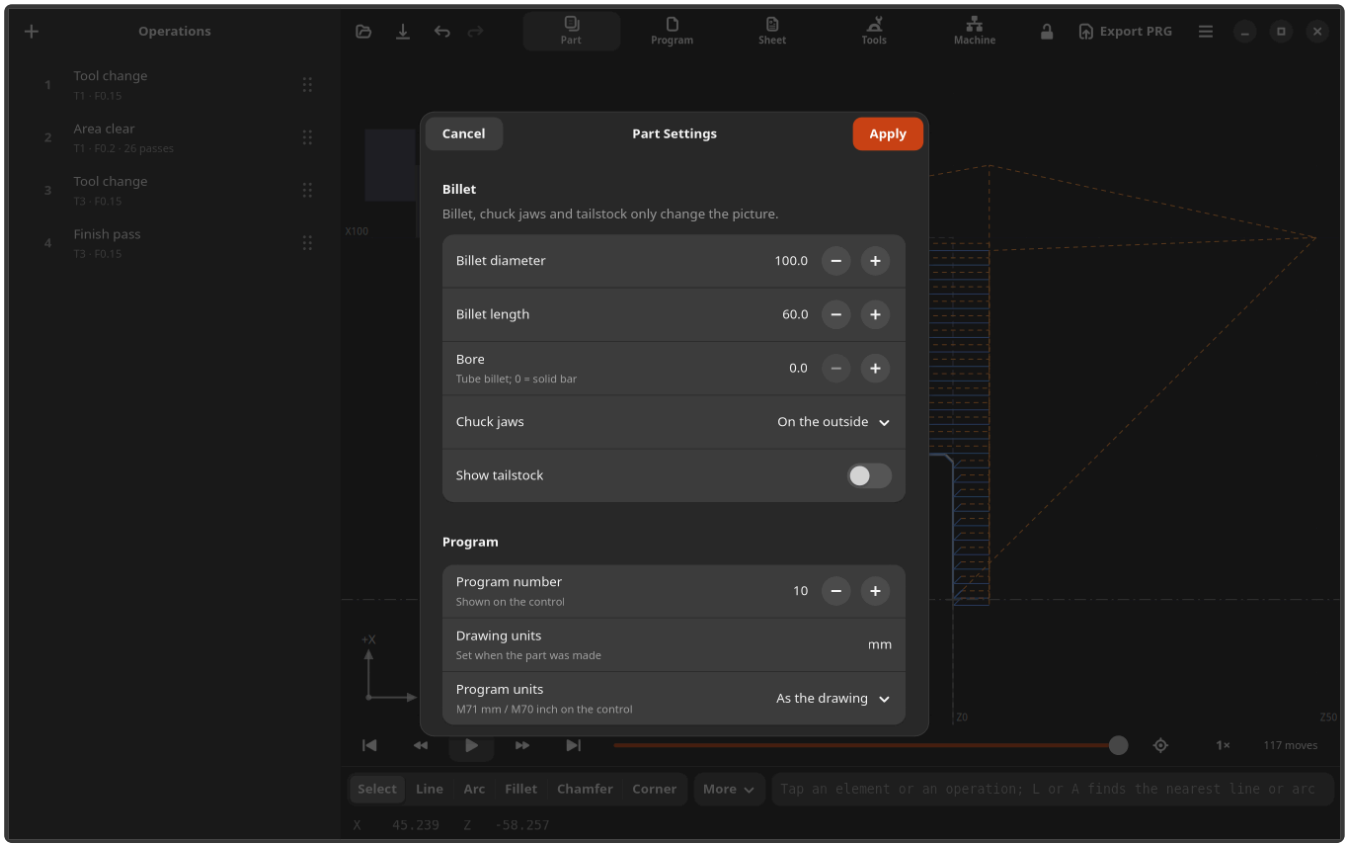
Main menu ▸ **Turn Part Around…** makes the setup for the other end of the part. The drawing is turned end for end about the middle of the part's length, so the back face is the new Z0 and the old front face lands at Z−length; arcs change direction. The new setup has no operations, the billet as long as the part, and the next program number, so both programs fit in the control. Setup 2 is kept in the same .alcam as setup 1. A Setup 1 / Setup 2 switch appears above the operations list; each setup has its own drawing, billet, operations, program number and undo history, and Export PRG (suggesting NAME2.PRG), the sheet and Send work on the setup shown. Turning around again replaces setup 2 after asking; Remove This Setup… (main menu) takes it out.

### Program number and units

The program number shown on the control (#132) is set in Part Settings (main menu; 1 to 9999); Z, feed and pitch are written doubled for the T macros automatically. Programs are written in the part's units (M71 metric, M70 inch); the control must be set to the same units.

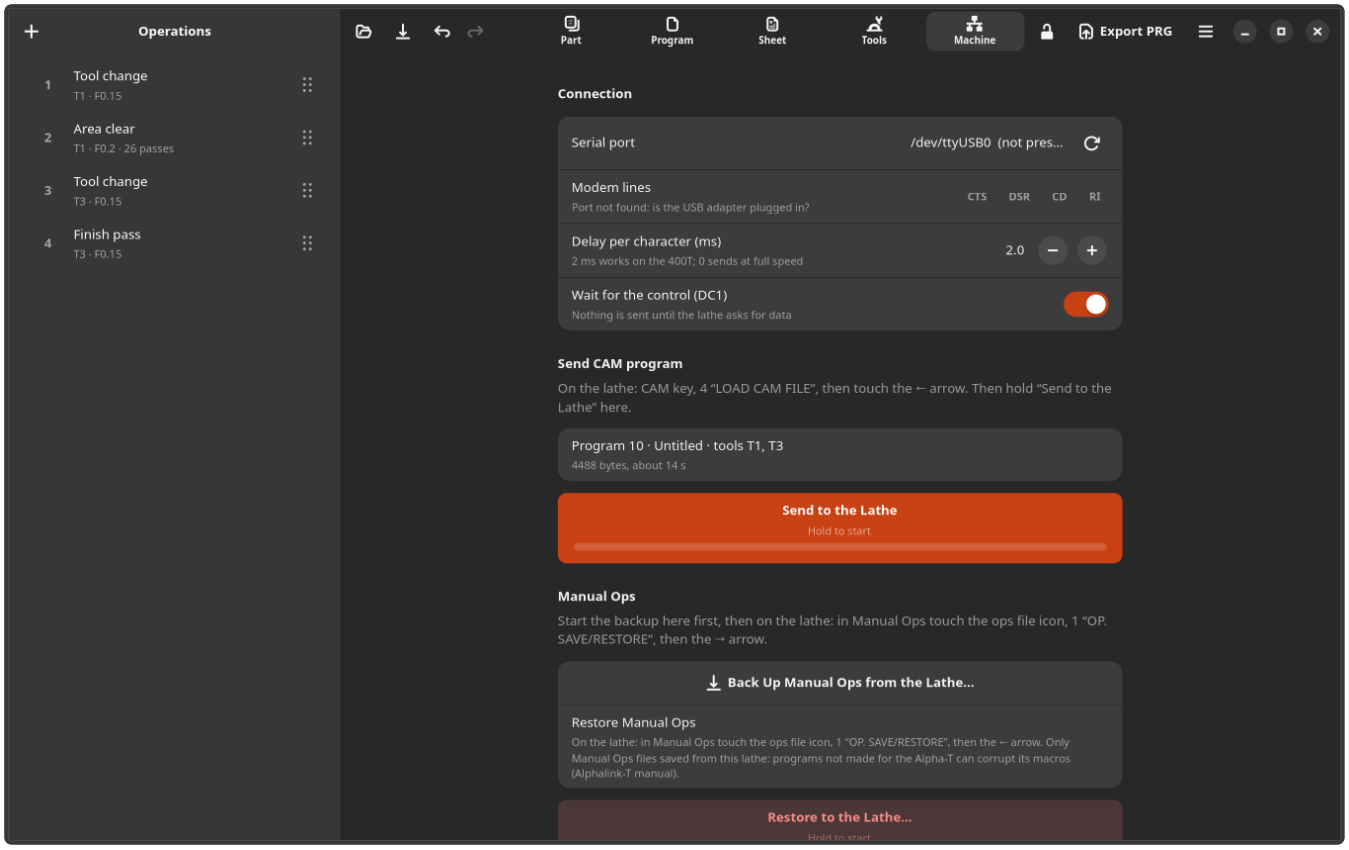
**Program units** in Part Settings writes the program in mm (M71) or inch (M70) whatever units the part is drawn in, like ALPHA-T's "Select Units To Store Output In". Lengths, feeds and pitches are converted, exactly as ALPHA-T does; the .SD1 stays in the drawing's units. The default is the drawing's units.

*Note:* unlike ALPHA-T there is no 8-character name limit; the exported file name is up to you. Export suggests the part's name cut to 8 characters, for the lathe's sake, and opens the folder used last.




## Sending to the machine

Everything is on the **Machine** page.




## Connection

| Setting | Meaning |
|---|---|
| Serial port | the USB serial adapter, usually /dev/ttyUSB0; the refresh button looks again |
| Modem lines | CTS, DSR, CD and RI as the lathe sets them; the lathe signals ready on them |
| Delay per character | 2 ms works on the 400T; 0 sends at full speed |
| Wait for the control (DC1) | nothing is sent until the lathe asks for data |

The line is 9600 baud, 7 data bits, even parity, 2 stop bits, with XON/XOFF, as ALPHA-T uses. The cable is the one in the Alphalink-T manual: DB9 pins 2-2, 3-3, 5-7, with the bridges in the machine plug (4-5 and 6-8-20).

**Bluetooth serial adapter.** Pair the adapter in the system's Bluetooth settings; the port list then offers it as "name (Bluetooth)" (the app connects over RFCOMM itself) and also shows any rfcomm device. Over Bluetooth the PC's port settings do not reach the line, so:

- set the adapter itself to 9600 baud, 7 data bits, even parity, 2 stop bits, and its own flow control to none, so that the lathe's DC1 and DC3 pass through to the app;

- an adapter that only frames 8 data bits goes to 9600 8N2 with **Parity from the app** on: the app adds the even parity bit, which gives the same frame on the wire;
- use a delay per character of 5 ms or more: Bluetooth adds latency, and the control raises alarm 087 if more than 10 characters arrive after its DC3;
- there are no modem lines over Bluetooth (the row is hidden), and the adapter needs its own power: the lathe's port does not supply it. Wired like a PC port (DB9 male, DTE), it takes the same cable.

### Load a CAM program

1. On the lathe press the CAM key, choose **4 LOAD CAM FILE** and touch the ← arrow.
2. In the app hold **Send to the Lathe** until it fills, check the summary (program number, tools, size) and confirm. From the keyboard, hold Space or Enter on it for the same 1.5 s; the confirmation opens when you let go, and Alt+S confirms, since Cancel is the default.
3. A window shows the bytes sent and can cancel. The program number then shows on the control.

*Note:* load only programs from ALPHA-T or this app (they are the same); others can damage the T macros. The standard control holds about 16 KB of CAM program. The app never starts the machine.

### Manual Ops backup and restore

- **Back up**: tap **Back Up Manual Ops from the Lathe…** first; then on the lathe in Manual Ops touch the ops file icon, choose **1 OP. SAVE/RESTORE** and touch →. Save the file.
- **Restore**: on the lathe the same menu and ←, then hold **Restore to the Lathe…** (or hold Space or Enter on it; Alt+R confirms) and choose the file. Only files saved from this lathe.

### Cable test

Unplug the cable from the lathe, bridge pins 2 and 3 of the adapter and tap **Run Loopback Test**. It shows whether the adapter and port work. Every transfer is logged in ~/.local/state/alphalink-cam/logs.

## Tool library

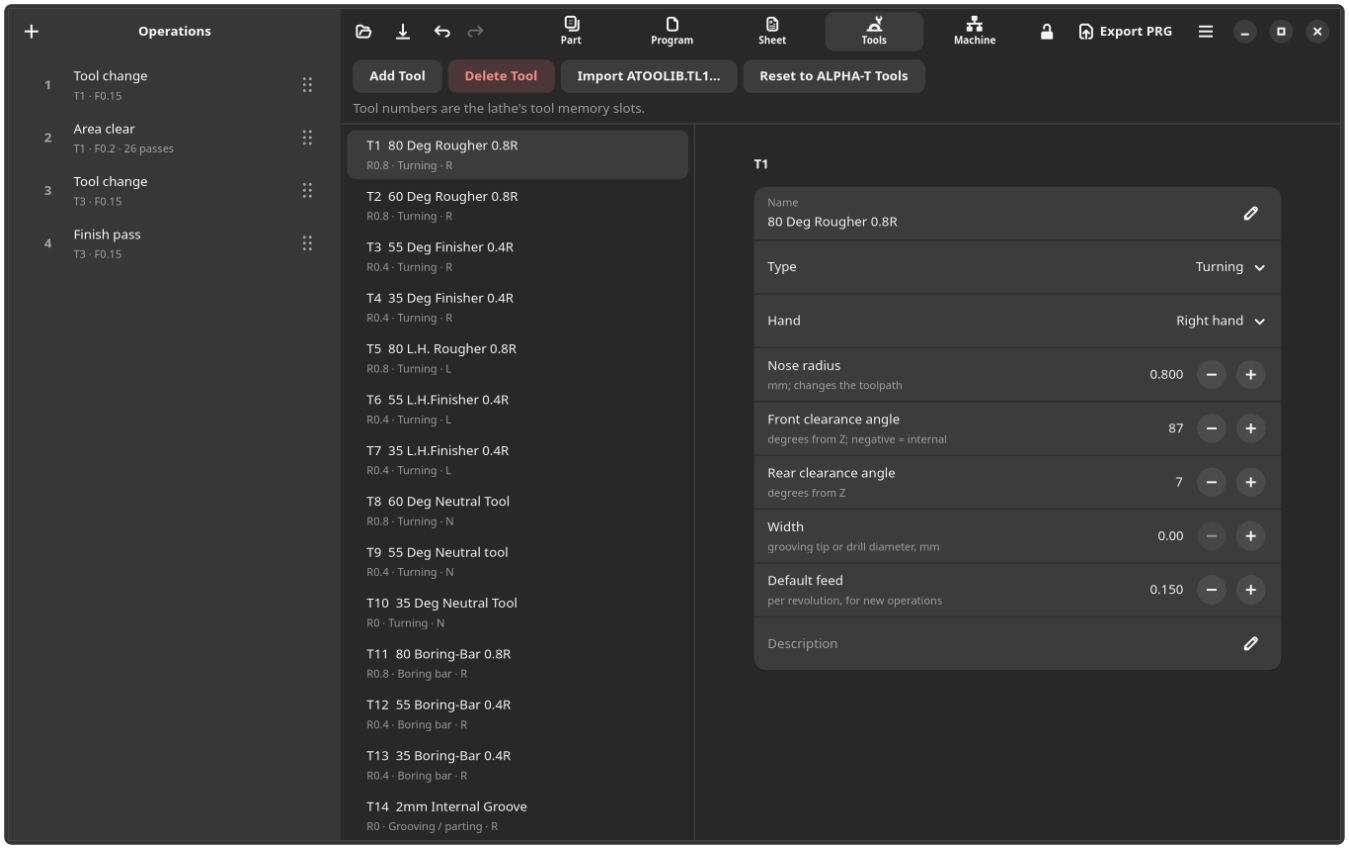
The **Tools** page replaces ALPHA-T's tool library editor (ATTOOLS). The tool numbers are the control's tool memory slots: an operation's **Tool** must be the slot that holds that tool's offsets.

| Button | What it does |
|---|---|
| Add Tool | a new tool in the next free slot |
| Delete Tool | the bin button in the selected tool's group header: removes the tool after asking |
| Import Tool Library… | reads an ALPHA-T tool library (ATOOLIB.TL1) |
| Reset to ALPHA-T Tools | in the ⋮ More Tool Actions menu: back to ALPHA-T's standard library (32 tools) |

Each tool has:

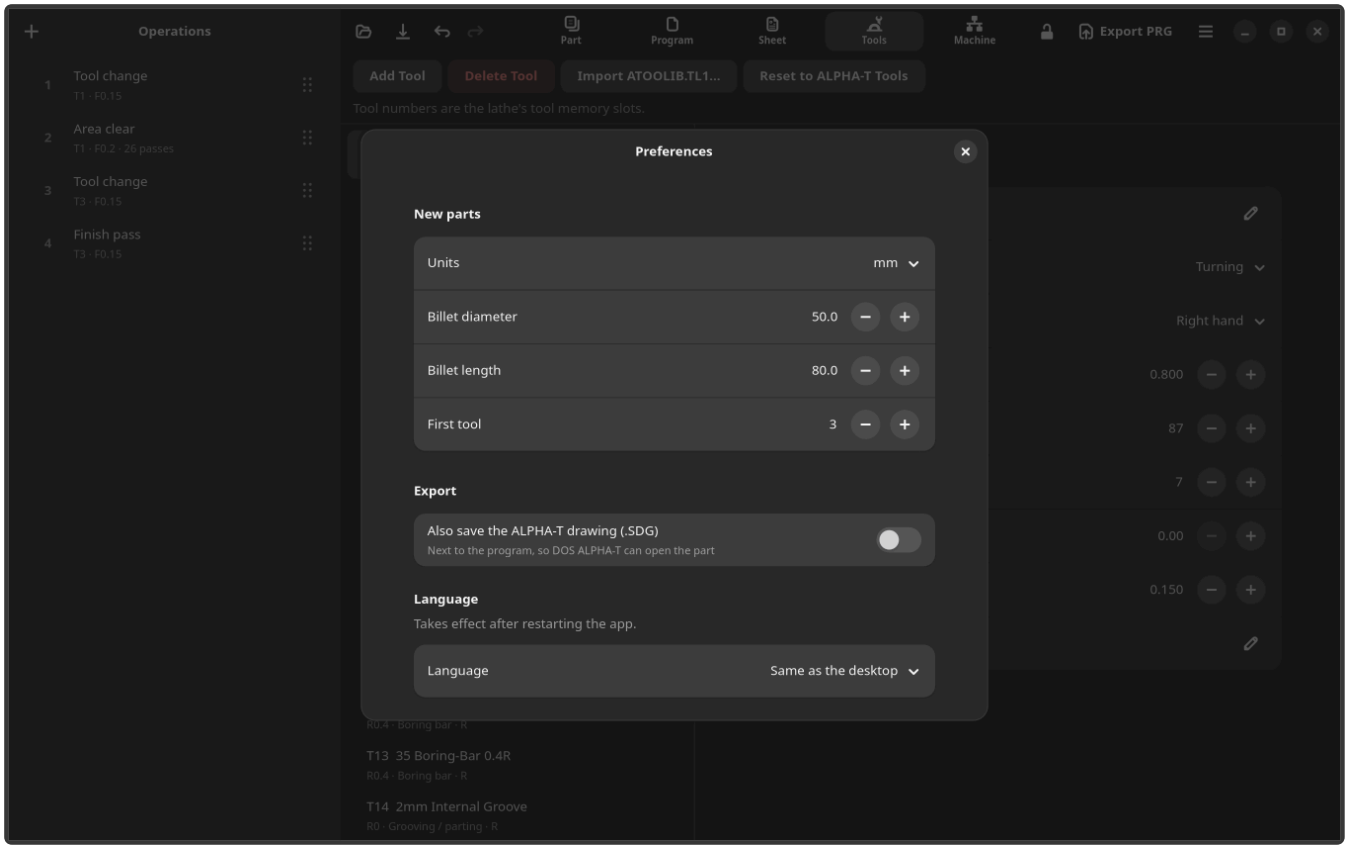
| Field | Meaning |
|---|---|
| Name | as shown in the editor and on the setup sheet |
| Type | turning, boring, grooving, threading, drilling |
| Hand | right, left or neutral |
| Nose radius | changes the toolpath (nose compensation) |
| Front / rear clearance angle | degrees from Z; ALPHA-T uses them to keep clear of the part |
| Width | grooving tip width or drill diameter |
| Default feed | the feed new operations start with |
| Description | free text |
| Surface speed | m/min, for the cycle time estimate; 0 = a default for the tool type |
| Spindle limit | rpm for this tool; 0 = the machine's limit |

The library is kept in ~/.config/alphalink-cam/tools.json; a part can also carry its own tools.




# Preferences, input and keyboard

**Main menu ▸ Preferences** (Ctrl+,): units, billet size and first tool for new parts, and the language (English, Ukrainian, or as the desktop; after a restart). Other main-menu items: **Fit Part**, **Show Grid**, **Part Settings**, **Help** (F1); the **Training Manual** and **Reference Manual** buttons are at the top of the Help window. **Preferences ▸ Input ▸ Input layout** switches between Automatic, Pointer and Touch layouts; Automatic follows the last two presses: finger, or mouse and touchpad. Preferences ▸ Cycle time sets the spindle limit (2500 rpm), the rapid rate (6000 mm/min) and the time a tool change takes (30 s) for the estimate.




## Touch, pen and mouse

- One finger drags the view (a quick flick keeps it moving), two fingers pinch to zoom, a double tap fits the part (again goes back).
- Mouse wheel pans up and down, Shift+wheel sideways, Ctrl+wheel zooms; a middle-button drag pans too.
- In touch layout the bars get bigger, a keypad button opens a number pad for X and Z, and a finger on any number field opens the keypad instead of the system keyboard.
- **Pen**: it draws and picks precisely, like a mouse, and does not switch the layout. After a pen press the zoom buttons and the **Keypad** button show, number fields open the keypad, and a long press with the pen opens the drawing's menu; the barrel button is a right click.
- **Touch Lock** (padlock in the header) ignores touches, for example while coolant runs; hold it to unlock by touch. It also turns on during transfers.
- A faint grid follows the zoom; every fifth line is labelled (X as a diameter).

## Keyboard summary

| Keys | Action |
| --- | --- |
| Ctrl+N / O / S / Shift+S | new, open, save, save as |

| Keys | Action |
|---|---|
| Ctrl+E | export PRG and SD1 |
| Ctrl+Z, Ctrl+Shift+Z or Ctrl+Y | undo, redo |
| Ctrl+0, Ctrl++, Ctrl+- | fit, zoom in, zoom out |
| Esc | end the line or command, cancel picking, deselect; keyboard back to the drawing; also clears the messages at the bottom |
| Delete | erase the selected element, anywhere on the Part page outside text fields |
| F1 | help; the Training Manual and Reference Manual buttons are at its top |
| Ctrl+? | all keyboard shortcuts, including the ALPHA-T letters |
| Ctrl+W | close the window |
| Ctrl+Q | quit |
| ALPHA-T letters | see Drawing commands (CAD) |
| Ctrl+F | search the program; N10 goes to block 10 |
| Shift+F10, Menu key, right click | the drawing's menu: Edit Values, Delete, Zoom Here, Fit Part |
| Alt+1 … Alt+5 | Part, Program, Sheet, Tools and Machine pages |
| Ctrl+PgUp / Ctrl+PgDn | previous / next page |
| Alt+↑ / Alt+↓, Ctrl+D, Delete | move the operation up / down, duplicate it, delete it (operations list or editor focused) |
| F5 | simulation: play / pause |
| Ctrl+[ / Ctrl+] | simulation: previous / next pass |
| Alt+Home / Alt+End | simulation: first move / show all |
| PgUp / PgDn on the simulation slider | move a tenth |
| Ctrl+= / Ctrl+− / Ctrl+0, Ctrl+wheel, pinch | Program page: text size (remembered) |
| Ctrl+arrows, middle-button drag | pan the drawing (Ctrl+arrows by an eighth of the view); the view also follows the cursor when the arrow keys move it past the edge |

| Keys | Action |
|---|---|
| [ / ] | select the previous / next element along the profile |
| Space (picking a contour) | add or remove the element at the cursor |
| Space or Enter, held 1.5 s | Send to the Lathe / Restore to the Lathe…: the confirmation opens when you let go; Alt+S / Alt+R confirms (Cancel is the default) |
| Keypad: digits, ., -, Backspace, Tab or ,, i, Enter | type, change the sign, delete, next field, switch ABS / INC, ✓ |

## Problems and support

| You see | Cause | What to do |
|---|---|---|
| Export PRG is greyed out, a banner at the top | an operation cannot be made into a program | read the banner: it names the operation and the reason |
| "… AWAY: ALPHA-T itself …" | AWAY on area clear, face clear or threading | keep **Towards the chuck** on; ALPHA-T's own AWAY programs cut past the shoulder, make one pass only or start inside the thread |
| "… (no sample)" | an option ALPHA-T has but no sample shows yet | use the default; samples are being made |
| "contour is not connected at X… Z…" | a gap or a wrong element in the contour | pick the contour again, in cutting order |
| Opening an .SDG gives no operations | the .SD1 is not next to it | **Choose .SD1…** in the message |
| Nothing arrives at the lathe | port, cable, or the lathe not waiting | check the port, the LOAD CAM FILE screen and its arrow, then the loopback test |
| The control shows the wrong tool or offsets | tool number is not the tool memory slot | set the operation's **Tool** to the slot |

| You see | Cause | What to do |
|---|---|---|
| The window is cut off | too narrow | below 1270 px the operations list hides behind the Operations button, below 1010 px the editor slides in over the drawing; widen the window |
| Letters type into the entry | the X,Z entry has the keyboard | press Esc |
| "… N5nn would be used twice …" | the program is too long for ALPHA-T's label numbers (from about 25 operations): a restart could start in the wrong operation | split the part into two programs |

Transfer logs are in ~/.local/state/alphalink-cam/logs; settings in ~/.config/alphalink-cam. The program's own help is under F1.